

LLM for Teens

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■ The Problem We Are Solving

In Session 4, you learned MCP. MCP gives one agent the ability to use tools — search a database, read a file, call an API. That works well when one agent can handle the whole task.

But some tasks are too big for one agent. Planning a trip requires searching flights, checking a calendar, booking a hotel, and sending a confirmation. One agent trying to do all of that gets messy — too many tools, too much context, too many decisions.

The solution: use more than one agent. Each one is a specialist. And the specialists need a way to talk to each other. That is A2A.

What A2A Is

- A2A stands for Agent2Agent. It is an open protocol created by Google that lets different AI agents communicate and collaborate, regardless of the framework or vendor they were built with.
- If MCP is the "USB-C for tools," A2A is the "HTTP for agent collaboration." It lets agents talk to each other as peers, share context, and coordinate work.

A2A vs MCP

- This is the key distinction for the session.
- MCP connects an agent down to a tool. The tool does one job and returns a value. The agent stays in charge.
- A2A connects an agent across to another agent. That peer can reason, hold its own state, ask clarifying questions, and use its own tools.
- So:
 - MCP gives an agent its capabilities.
 - A2A lets that equipped agent collaborate with others.
 - MCP is vertical. A2A is horizontal. Both are needed in a real system.

How A2A Works

- The flow is simple:
 - Step 1: An agent (the client) identifies a task it cannot do alone.
 - Step 2: It discovers a remote agent that can help. Each A2A agent publishes an "agent card" — a small description of what it does and what it can accept.
 - Step 3: The client sends the task to the remote agent.
 - Step 4: The remote agent accepts the task, does the work (it may use its own MCP tools), and returns the result.
 - Step 5: The client continues, using the result.
- The two agents don't need to share code or frameworks. Agent 1 could be built with Gemini

Multi-Agent Systems

- Once you have A2A, you can build a team. For example:
 - A Researcher Agent that searches for information.
 - A Writer Agent that turns the research into a draft.
 - A Reviewer Agent that checks the draft for errors.
 - They share tasks through A2A, divide the work, and solve bigger problems than any one agent could alone.

Why This Matters

- Single agents hit limits fast. Too many tools confuse them. Too much context slows them down. Too many steps cause errors.
- Splitting the work across specialists keeps each agent focused. Each one has fewer tools, a smaller context, and a clearer job. The result is a system that is easier to build, easier to debug, and more reliable.

Assignment

- In this assignment, you will build a small multi-agent system using A2A (Agent2Agent). The goal is to see how two agents work as a team, each with its own job, and pass results between them using a standard protocol.
- Dataset
 - You must use the following dataset: Kaggle Bank Customer Complaints
- Tasks
 - 1. Build a Researcher agent that summarizes complaints for a given category. Run it on port 5001.
 - 2. Build a Writer agent that turns the summary into a short report. Run it on port 5002.
 - 3. Connect with an A2AClient. Send a category to the Researcher, then pass the result to the Writer.
 - 4. Show the full flow, including the server logs.
 - 5. Add a third agent (Reviewer) on port 5003 that checks the report.

Summary

- ✓ What A2A is
- ✓ A2A vs MCP
- ✓ Agent-to-agent communication
- ✓ Multi-agent systems
- ✓ Real demo: two agents working as a team

Thank You

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